



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS BACHELOR OF
COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOURS

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

PROJECT TITLE:

GROUP 14

RECRUITX - EVENT-DRIVEN JOB RECRUITMENT & APPLICATION PROCESSING
SYSTEM

INDIVIDUAL LOGBOOK

PREPARED FOR:

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PREPARED BY:

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1. Individual Logbook Report

RecruitX - Event-Driven Job Recruitment & Application Processing System

1.1. Team Information

Item	Details
Name	Maryam Musfirah Binti Ariff Hazizan
Matric No	S75922
Course	Bachelor of Computer Science (Software Engineering) with Honours
Subject	CSE3433 Software Architecture
Project Title	RecruitX – Event-Driven Job Recruitment & Application Processing System
Group Number	Group 14
Architecture Style	Event-Driven Architecture (EDA)

1.2. Team Members

Name	Matric No	Role
Maryam Musfirah Binti Ariff Hazizan	S75922	Wrote backend testing scripts, validated message consumer logs, and compiled final evaluation benchmarks.

Ainaa Nadhirah Binti Asmadi	S75246	Recruiter Portal Development, Candidate Management
Nur Alin Hayani Binti Ahmad Syukri	S75034	Designed frontend UI mockups and client-side view routers for the portals.

2. Weekly Progress Log

2.1. Week 4: Requirement Analysis and Service Identification

Category	Details
Activities	- Discussed project scope and selected RecruitX as the project topic. - Analysed problems in traditional recruitment processes. - Identified candidate and recruiter user roles. - Identified key events required in the system.
Individual Contributions	- Led the team's problem diagnostic analysis. - Detailed how manual communication gaps lead to top-talent dropouts and drafted the core business goals to overcome these delays.
Challenges	Pinpointing exact operational entry points where automation can replace manual HR intervention seamlessly.
Action Plan	Gather formal academic literature on Event-Driven structural patterns to justify our design model.

2.2. Week 5: System Requirements Specification

Category	Details
Activities	Defined the baseline Functional (FR) and Non-Functional Requirements (NFR) for the entire architectural pipeline.
Individual Contributions	- Authored the exact specifications for the background automation triggers (specifically focusing on automated calendar coordination and immediate candidate confirmation criteria). - Defined NFR targets for reliability and fault tolerance.
Challenges	Differentiating between standard procedural interactions and true event-based architectural dependencies.
Action Plan	Blueprint the high-level System Context Diagram with the team to establish outer system boundaries.

2.3. Week 6: Interaction Modeling and Architecture Blueprints

Category	Details
Activities	Created detailed multi-tier UML architecture models.

Individual Contributions	● Partnered with peers to refine the concurrent interaction sequences for asynchronous data handling.
Challenges	● Accurately reflecting decoupled message queuing workflows without implying direct, tightly-coupled component calls.
Action Plan	● Coordinate with the database lead to ensure notification structures map correctly to relational data tables.

2.4. Week 7: Schema Engineering and Notification Mapping

Category	Details
Activities	● Finalized internal relational data mapping rules and messaging properties.
Individual Contributions	● Defined structural attributes for the notifications and interviews tables within the central MySQL database. ● Mapped out how raw payload data from the message broker would populate historical candidate alerts.

Challenges	● Preventing relational update conflicts when independent background consumers write to tables simultaneously.
Action Plan	● Begin constructing the Java backend logic for the automated notification consumer.

2.5. Week 8: Notification Service Infrastructure

Category	Details
Activities	● Established data pipeline triggers and initialized notification alert modules.
Individual Contributions	● Programmed the backend core logic of the NotificationConsumer using Java Servlets. ● Developed internal routing rules to catch structural alert updates and push simulated feedback states.
Challenges	● Managing state exceptions safely if an email service system simulation times out unexpectedly.
Action Plan	● Shift technical focus to constructing the backend scheduling engine and calendar hook-ins.

2.6. Week 9: Interview Scheduling Engine Development

Category	Details
Activities	● Formulated automated workflow modules for managing recruiter interview requests.

Individual Contributions	- Engineered the core processing backend for managing InterviewRequested message payloads. - Built functional handling utilities that automatically pair selected dates with generated virtual meeting spaces.
Challenges	Keeping scheduling states uniform across multiple background steps without introducing operational lags.
Action Plan	Construct comprehensive validation test cases to monitor consumer performance under simulated pressure.

2.7. Week 10: Testing Script Engineering & Consumer Logging

Category	Details
Activities	Designed backend validation scripts to inspect asynchronous background operations.
Individual Contributions	- Created automated internal script assertions to keep track of event deliveries. - Verified that incoming status modifications immediately triggered the target consumers without human steps.
Challenges	Tracking down and debugging silent database execution drops occurring within deeply nested background threads.
Action Plan	Collaborate on end-to-end component integration tests to assess multi-tenant workflow reliability.

2.8. Week 11: End-to-End Workflow Integration

Category	Details
Activities	• Unified decoupled modules into a singular functioning system stack.
Individual Contributions	• Conducted system-wide tracking validation by connecting the notification alerts directly to live pipeline status changes. • Monitored the background console execution to track how event records populated sequentially.
Challenges	• Managing sporadic event transmission order errors during high-volume testing phases.
Action Plan	• Compile final system metrics and assemble technical evaluation documentation for the final submission.

2.9. Week 12: Architecture Evaluation and Report Synthesis

Category	Details
Activities	• Ran final platform validation checks, resolved architectural edge-cases, and synthesized the final report.

Individual Contributions	• Led the architectural evaluation and benchmarking phase. • Wrote the final performance summary detailing how the Simulated RabbitMQ ecosystem handles temporary service failures through localized data isolation.
Challenges	• Balancing technical details with clear explanations inside the final project documentation.
Action Plan	• Perform a final review of the project materials, clean the Git repository, and prep for group assessment.

3. GenAI Usage Record

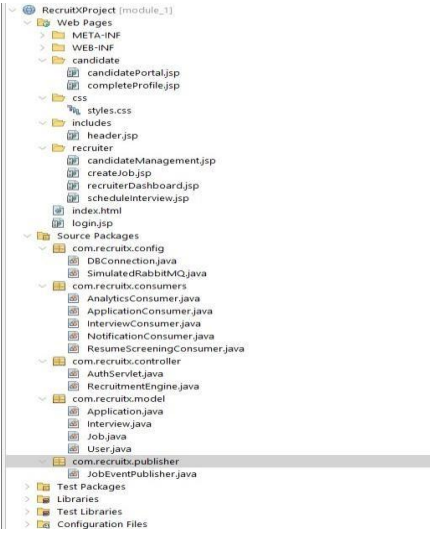
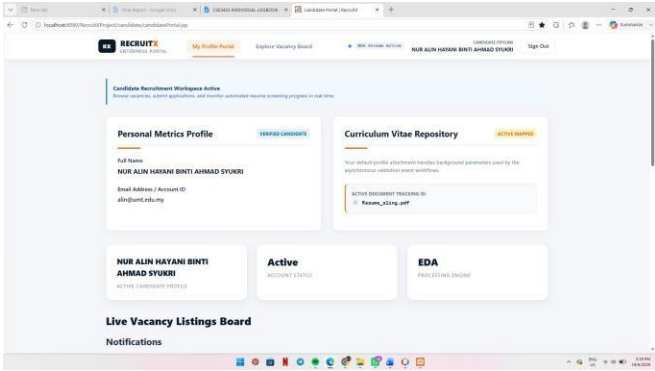
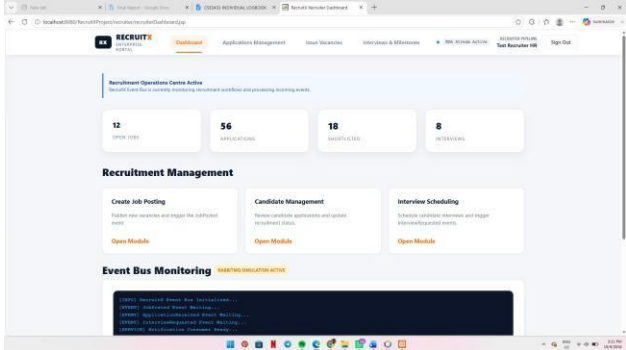
Week	GenAI Tool	Purpose	Prompt Used	Output Summary	Reflection and Adaptation
Week 5	ChatGPT	Refine architectural specification vocabulary.	"Draft precise non-functional availability and reliability metrics for a distributed message-driven human resource pipeline."	Outlined performance benchmarks and transactional retry standards.	Extracted specific availability metrics to strengthen our system NFR definitions.

Week 6	ChatGPT	Map abstract external service interfaces clearly.	"How should third-party email platforms and calendar engines be drawn inside an architectural context model?"	Provided structured layouts using distinct system boundaries.	Directly guided the final formatting of the outer context drawing blocks.
Week 10	ChatGPT	Optimize logging parameters for backend exceptions.	"How do I print safe execution tracking traces inside nested multi-threaded consumer classes?"	Showed structured logging examples using standardized code formats.	Used to construct the custom console output visible inside the final event processing interface.
Week 13	ChatGPT	Formulate structural project reflections.	"Draft an architecture review highlighting data isolation advantages when backend components fail."	Provided conceptual summaries on fault containment patterns.	Tailored this structure into our architectural summary to clearly describe our error handling.

3.1. Reflection

Expert Generative AI was used for formulating complex technical requirements and documenting the same. Though it provided a great framework to log the errors in code as well as keep track of it, all the system testing scenarios, requirement collection, and validation of system code were done manually to satisfy our exact project needs.

4. Evidence Summary

Evidence	Description
	The project structure demonstrates the separation of responsibilities through controllers, consumers, publishers, models, and configuration classes. This structure supports maintainability and modular development.
	The Candidate Portal allows users to view available vacancies, manage their profiles, monitor application progress, and receive recruitment updates.
	The Recruiter Dashboard provides access to vacancy management, candidate management, interview scheduling, and event monitoring functions.

		id	event_type	payload	processed_at
		1	ApplicationReceived	jobid:1,candidateid:3,resume:Resume_aling.pdf	2026-06-07 15:57:17
		2	ApplicationReceived	jobid:1,candidateid:3,resume:Resume_aling.pdf	2026-06-09 03:18:24
		3	ApplicationStatusUpdated	applicationid:1,status:ACCEPTED,recruiterid:1	2026-06-09 03:28:35
		4	ApplicationStatusUpdated	applicationid:2,status:SHORTLISTED,recruiterid:1	2026-06-09 03:35:37
		5	ApplicationStatusUpdated	applicationid:2,status:INTERVIEW,recruiterid:1	2026-06-09 03:50:01
		6	ApplicationReceived	jobid:1,candidateid:6,resume:Resume_aling.pdf	2026-06-09 14:45:10
		7	JobPosted	title:lead backed software engineer	2026-06-09 15:05:29
		8	ApplicationReceived	jobid:1,candidateid:6,resume:Resume_aling.pdf	2026-06-09 15:12:01
		9	ApplicationReceived	jobid:1,candidateid:6,resume:Resume_aling.pdf	2026-06-09 15:12:31
		10	ApplicationReceived	jobid:1,candidateid:8,resume:Resume_aling.pdf	2026-06-13 22:06:21
		11	ApplicationStatusUpdated	applicationid:6,status:SHORTLISTED,recruiterid:7	2026-06-13 22:12:12

The event_logs table records all published events including JobPosted, ApplicationReceived, ApplicationStatusUpdated, and InterviewScheduled, demonstrating the operation of the event bus.

		id	application_id	fit_score	keyword_matches	status
☐		6	6	40	Basic Criteria Met	REJECTED
☐		7	7	85	Java, Software Architecture, SQL, Web Systems Mana	SHORTLISTE

The screening results table stores automated fit scores and screening outcomes generated by the ResumeScreeningConsumer service.

The interviews and notifications tables verify successful execution of interview scheduling and notification services within the event-driven workflow.

5. Final Individual Reflection

Development of RecruitX has helped me gain an expansive view on requirement planning for systems and software quality assurance in distributed software systems. Using Java Servlets to create interview scheduling and notification consumers helped me understand in depth the significance of event isolation in ensuring platform availability. Managing the issues of validation of decoupled code paths and asynchronous data tables was a rewarding experience for me.